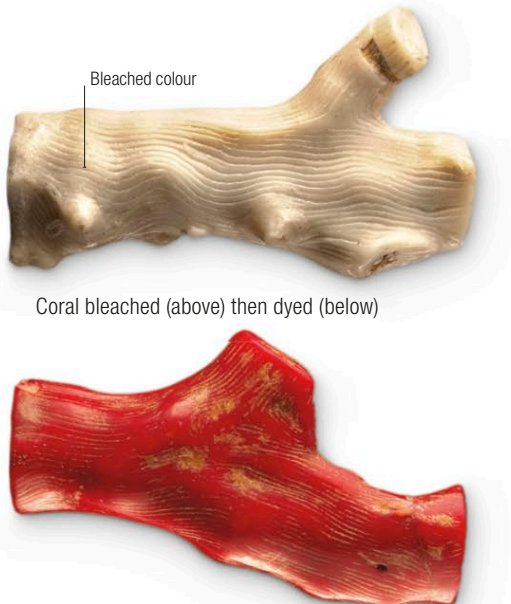


Enhancing gems

Many gemstones undergo treatment to enhance their natural characteristics in addition to the standard cutting and polishing processes that a lapidary will carry out. Gem sellers are expected to disclose any additional treatments a stone may have undergone. This is not only because a gem's value is based around its natural characteristics, but also because some treatments can also affect a gem's durability, or wear off. For example, the artificial coating of topaz gems (see right) can become scratched and wear off.



Bleached colour

Coral bleached (above) then dyed (below)

Dyeing, bleaching, and staining

The staining and dyeing of gems is widespread. Slices of agate are routinely dyed with vivid colours such as blue or red – often using ordinary household cloth dyes, although there are also dyes specifically for stones. If a dye comes off a gem onto the hands, it is a sign of poor quality.



Enhanced colour

Collection of irradiated gemstones in various colours

Irradiation

A gemstone's colour can be altered by irradiation – bombardment with neutrons, gamma rays, ultraviolet light, or electrons, which is often followed by heat treatment. Most blue topaz on the market is irradiated and heat-treated colourless topaz.



Tanzanite surface-coated to deepen and improve colour – before (left) and after (right)

Surface coating

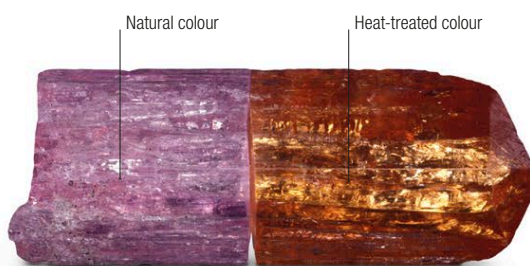
Very thin coatings of gold, silver, and other metals can be applied to gems to alter their colour or reflectivity. This can be seen in varieties such as “mystic” topaz and “aqua aura” quartz. While attractive, these coatings soon wear off.



Gemstones after being repaired by filling

Filling, coating, and reconstruction

Some gemstones with cracks are subjected to fillers other than the oiling described for emerald. Fillers can be glass, resins, plastic, or waxes, and can be coloured to match the gemstone. Heat, pressure, or solvents are used to fuse together small pieces.



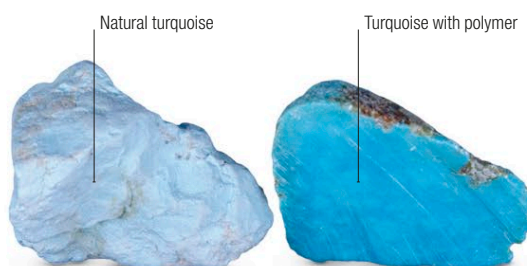
Natural colour

Heat-treated colour

Imperial topaz crystal sawn in half: the crystal on the right was heated. Both colours are highly desirable

Heat treatment

This is among the oldest forms of gemstone enhancement – zircons have been heat-treated to change their colour for at least a millennium. Today, the process is used to change colour or conceal inclusions.



Natural turquoise

Turquoise with polymer

Porous turquoise impregnated with a wax or polymer substance to colour and stabilize the material

Oiling

An old method of treating cracked emerald is simply to soak the stone in oil. This fills the cracks and makes the stone appear of a higher quality; however, it can give the gem an oily feel. Turquoise can also be soaked in polymer to improve colour.



Laser-drilled table facet – the drill holes have created a cleavage crack, worsening the flaw

Laser drilling

Diamond is combustible, a property that allows infrared lasers to drill tiny holes into the material to reach flaws and inclusions. Once reached, inclusions can be dissolved, and flaws can be filled with epoxy resins.